

Phase 1: Core Logic & Automation

Focus: Shaking off the rust and building muscle memory with syntax.

Video	Core Topic	Hands-On Project / Goal
Video 1	Variables & Data Types	Set up environment, understand memory allocation.
Video 2	REPL & Smart Logic	Build quick terminal logic testers.
Video 3	Python Operators	Mathematical and logical operations mapping.
Video 4	Automation Basics	Script to send Birthday & Anniversary emails automatically.
Video 5	Conditional Statements	Build a Zodiac App & basic terminal games.
Video 6	Strings Manipulation	Password Validation logic (crucial for future web auth).
Video 7	Loops (For/While)	Automating repetitive tasks.

Phase 2: Data Structures & State Management

Focus: How to store, retrieve, and manipulate data efficiently.

Video	Core Topic	Hands-On Project / Goal
Video 8	Master Lists	Basic UI integration with Streamlit.
Video 9	API & External Libraries	Build a Voice + Text Virtual Assistant (Jarvis).
Video 10	Tuples	Voice to PDF Report Generator.
Video 11	Dictionaries	Key-value pairing (mimics JSON, essential for web APIs).
Video 12	Application State	Build a Pomodoro Study Break Timer.
Video 13	Bulk Processing	Voice to PDF Bulk Report Generator.
Video 17	Sets	Document Scanner / Image to Text tool.

Phase 3: Advanced Architecture (Stop the Playlist Here)

Focus: Writing clean, modular, professional code.

Video	Core Topic	Hands-On Project / Goal
Video 18	Functions	Modular ATM System.
Video 19 & 20	Recursion	Logic puzzles and technical interview prep.
Video 21	File I/O	Reading/writing local files and interacting with APIs.
Video 22 & 24	Object-Oriented Programming (OOP)	Classes, Objects, Constructors. <i>This is mandatory for web frameworks.</i>

Phase 4: The Web Dev Pivot (Post-Playlist)

Once you finish OOP in Video 24, abandon this playlist and pivot to these technologies to build actual web applications.

Target Technology	Why You Need It	Next Steps
Django or Flask	The actual frameworks used to build Python web servers.	Find a dedicated tutorial for building a CRUD (Create, Read, Update, Delete) web app.
SQLite / PostgreSQL	Storing user data, passwords, and app information permanently.	Learn basic SQL and how to connect a Python app to a database.
REST APIs	Allowing your Python backend to talk to front-end web interfaces.	Build a simple API that returns JSON data to a browser.